Romanized Hindi Is Harder Than Hinglish: A Scoring-Aware Audit of Small LLMs

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Abstract

Prior work reports lower LLM accuracy on Romanized Hindi than on Romanized Hinglish versions of the same translated GSM8K questions, and shows that translation errors and answer extraction can skew multilingual math gaps. Building on this evidence, we audit small open LLMs on newly drafted, human-reviewed math triplets in English (E), Romanized Hindi (R) and Roman-script Hinglish (C), with code-computed gold and a frozen, scoring-aware protocol. On permanently excluded fixtures, a plausible answer-format instruction made 10/36 Qwen3-4B outputs fail strict scoring and pushed 8/12 R explanations into Devanagari; after one disclosed prompt revision both were zero, and format compliance across seven models was strongly model-specific. In two separately reviewed, pre-registered cohorts, never pooled, R is the lowest-scoring form for all three backbones. On the primary 70-family cohort, Qwen3-4B scores 94.3/85.7/94.3% on E/R/C, Qwen3-8B 98.6/91.4/98.6% and Llama-3.2-3B 94.3/57.1/70.0%; a 24-family cohort shows the same direction, with Qwen near ceiling. Both Qwen models have equal observed aggregate accuracy on E and C. R errors are not concentrated in items with Hindi operation words. In a pre-registered fine-tuning follow-up on 100 held-out families (Qwen3-4B QLoRA, three seeds), training on Hinglish rather than Romanized Hindi lowered Hinglish accuracy by 4.3 points $[-8.0, -0.7]$, and every fine-tuned variant fell far below the unchanged model.

1 Introduction

A model that answers "How many marbles does Aman have?" may fail on *Aman ke paas kitne kanche hain?* although the mathematics is identical. We hold the mathematics fixed and vary only the form.

The ordering we observe is not new. CodeMix-Tax (Sthānika, 2026) aligns the same 955 GSM8K questions across five forms and already reports Romanized Hindi below Romanized Hinglish, which is direct prior evidence for our main result. Code-mixed and Romanized math gaps appear elsewhere too (Yang and Chai, 2025; Sarvam AI, 2025; Das et al., 2026), and Peter et al. (2026) show that translation errors and answer extraction can skew multilingual math gaps. Building on this, we audit small open LLMs on newly drafted, human-reviewed E/R/C math triplets. We add: (1) **new triplets**, drafted rather than derived from GSM8K and written in natural Roman spelling rather than scheme transliteration, with human review; (2) **a scoring-aware frozen protocol**: on excluded fixtures a plausible final-answer instruction caused form-dependent scoring failures, and a seven-model compliance screen preceded model admission; (3) **two separately reviewed cohorts** whose direction is consistent; (4) **a pre-registered fine-tuning test** of whether Hinglish training helps Hinglish.

2 Related Work

Hindi and multilingual math. MGSM translates GSM8K (Cobbe et al., 2021) into ten languages, but not Hindi (Shi et al., 2023). GSM8K-Hi is a human-verified machine translation of GSM8K into Devanagari (Kamath et al., 2025), and HAWP has 2,336 Hindi arithmetic problems (Sharma et al., 2022). Anand et al. (2025) tune on Hindi and English math, excluding Romanized Hindi; mCoT (Lai and Nissim, 2024), MathOctopus (Chen et al., 2024) and XCOT (Chai et al., 2025) study multilingual math training.

Code-mixed and Romanized evaluation. CodeMixBench's Hindi–English GSM8K mixes Devanagari and Latin script (Yang and Chai, 2025); Indi-RomCoM varies Roman-

ized code-mixing intensity, without a separate monolingual-R condition (Das et al., 2026, preprint). Sarvam-M reports GSM8K accuracy falling from native-script to Romanized Indic input (Sarvam AI, 2025), and Romanized input also hurts health triage (Khullar et al., 2026).

Operation cues and spelling. Math solvers exploit shallow cues (Patel et al., 2021), and LLMs turn irrelevant phrases into operations (Mirzadeh et al., 2025). Romanized Hindi has no standard orthography (Bali et al., 2014; Roark et al., 2020). Token counts differ across languages (Petrov et al., 2023); we treat them only as correlates.

3 Data

Families and forms. A family is one problem with three renderings. **E** is English; **R** is Hindi phrasing in Roman script, with a borrowing list for numbers, units and established loanwords; **C** is Roman-script Hindi with English lexical material beyond that list. Six topics (comparison, money, fractions/percentages, ratios, rates/units, algebra) are covered. The original pilot drafted 30 families (five per topic), of which 24 form its cohort; Pilot-B adds 70 (12/12/12/12/11/11).

Drafting, code gold, blind verification. Gold is computed in code for every family. Claude (claude-opus-5-5) drafted Pilot-B; for the original 30, an author wrote the English statements and an OpenAI GPT model (via Codex) translated them into R and C. Claude and GPT each solved every Pilot-B rendering blind in a fresh context; no Qwen model ever drafted, solved or checked. Both solvers matched gold on 210/210 first-version and 15/15 revised renderings. This shows only that the text encodes the calculation; the solvers may share blind spots.

Human review: original cohort. Two humans (R01, R02) solved the mathematics independently before seeing gold; on the cohort's 72 renderings their answers agreed on 65, R02 matched gold on all 72, and the adjudicator (A01) confirmed gold for every cohort family. Language and equivalence review was by one human (R02) plus A01: R01's stage-2 judgments on 52 of the 72 cohort renderings were AI-prefilled and unverified, so they are excluded. Earlier review rounds were found non-independent and discarded. A01 decided 24 ACCEPT, 5 REVISE and 1 EXCLUDE (a near-duplicate); the cohort is the 24 families accepted without repair.

Human review: Pilot-B. One bilingual team member reviewed all of Pilot-B: 57 ACCEPT, 13 REVISE, 0 REJECT, with two equivalence failures. The same reviewer also reviewed a separate, not yet evaluated GPT-drafted (`gpt-6-astra`) test set; its Romanized Hindi was systematically over-formal (68/100 families needed repair, against 13/70 for Claude-drafted Pilot-B). This is descriptive: drafter is confounded with split, prompt and review timing; 62 of the 68 revisions targeted R.

C as insertional Hinglish. Review showed natural Hinglish to be mostly *insertional*: Hindi grammar with English nouns, conjunct verbs and relation words (*23 add karta hai*). Pilot-B C therefore requires English material beyond R's borrowing list but no English clause; the original cohort keeps the earlier clause-based definition, so the cohorts are reported separately.

Operation-word lexicon and keep/switch balance. An AI research assistant proposed 25 Hindi operation-word families with spelling variants, flagged as lexically ambiguous (*bhinn* "fraction"/"different"; *bhaag* "divide"/"run") or context-dependent (*kam*). One bilingual team member reviewed all entries (13 AI- or dictionary-assisted, then endorsed). On the original 30 drafts, a fixed matcher finds lexicon words in 27/30 R but only 10/30 C renderings, so a C advantage there could partly reflect easier vocabulary. Pilot-B counters this: C *keeps* R's Hindi operation word in 18 families, *switches* it to English in 17, and 35 are controls without one.

4 Protocol

Prompt and scoring. One English instruction (`test1-english-final-v2`) is used for every form. It asks for a short explanation in English and a final line beginning `FINAL:`, followed by an English unit only when the answer has one and by nothing after a pure number (Appendix A). It was revised once on excluded data (§5), then frozen. The primary endpoint is a frozen strict scorer (v1.0.0): exactly one terminal `FINAL:` line, an exact rational value, and a unit validated against predeclared aliases and conversions; missing, refused, ambiguous and truncated answers count as wrong. The secondary endpoint, `numeric_literal_v1`,

Model	Inv.	E/R/C	Dominant failure
Qwen3-1.7B$^{\times}$	36	0/0/0	instruction echo
Qwen3-4B†	0	12/10/12	2 R op-word misreadings
Qwen3-8B	0	12/12/12	none (1 R in Devanagari)
Llama-3.2-3B	2	12/9/9	value errors in R, C
Phi-4-mini$^{\times}$	14	6/4/11	inline FINAL; R repetition
Granite-3.3-2B$^{\times}$	36	0/0/0	commentary after FINAL
Gemma-3-4B$^{\times}$	–	–	no outputs (CUDA, FP16)

Table 1: Format screening on the excluded fixtures (prompt v2; strict-invalid of 36; strict correct of 12 per form). †primary; $^{\times}$excluded from the pilot. Phi-4-mini and Granite were excluded before the $\leq$3/36 rule was formalized; both exceed it.

Backbone	E	R	C
Pilot-B (primary), 70 families			
Qwen3-4B	94.3 [86.2, 97.8]	85.7 [75.7, 92.1]	94.3 [86.2, 97.8]
invalid	3	2	1
Qwen3-8B	98.6 [92.3, 99.7]	91.4 [82.5, 96.0]	98.6 [92.3, 99.7]
invalid	1	1	0
Llama-3.2-3B	94.3 [86.2, 97.8]	57.1 [45.5, 68.1]	70.0 [58.5, 79.5]
invalid	1	6 (5)	4 (2)
Original cohort, 24 families (no invalid outputs)			
Qwen3-4B	100.0 [86.2, 100.0]	91.7 [74.2, 97.7]	100.0 [86.2, 100.0]
Qwen3-8B	100.0 [86.2, 100.0]	95.8 [79.8, 99.3]	100.0 [86.2, 100.0]
Llama-3.2-3B	87.5 [69.0, 95.7]	62.5 [42.7, 78.8]	91.7 [74.2, 97.7]

Table 2: Strict accuracy (%) by form and cohort, with 95% Wilson intervals; the cohorts are never pooled. *invalid*: strict-invalid outputs (truncated in parentheses). No refusals occurred. Qwen3-4B is primary.

adopted before the pilot, checks whether the literal value on the sole FINAL line equals gold. It is reported beside, never instead of, the strict endpoint. Disagreement does not prove a format-only error, and agreement does not rule out formatting failures that both endpoints reject (cf. Peter et al., 2026).

Models. The primary backbone is Qwen3-4B (Qwen Team, 2025) at revision `1cfa9a72. . .`, with NF4 4-bit quantization (Dettmers et al., 2023), FP16 compute, thinking disabled, greedy decoding, a 1,024-token cap and a fresh context per prompt, on a Colab T4. Qwen3-8B and Llama-3.2-3B-Instruct are secondary, descriptive backbones under the identical protocol, in both cohorts. Both passed a rule declared before screening: at most 3/36 strict-invalid fixture outputs under the unchanged prompt and scorer, with no per-model tuning.

5 Development Findings (Not Results)

All evidence here comes from 36 AI-authored, unreviewed fixture prompts (12 families $\times$ E/R/C), **permanently excluded** from both cohorts and all later data; every rate is only an indication.

Format confound and v1→v2. Under the original instruction (`FINAL: <number or fraction> <unit when applicable>`), 10/36 Qwen3-4B outputs were strict-invalid, all `invalid_unit`; strict-correct counts were 9/7/9 on E/R/C. In 8, a unit word was appended to a dimensionless answer (`8 unitless`). With v2, 0/36 were invalid, the endpoints agreed on all 36, and counts were 12/10/12; as the fixtures also changed (one R repair, one alias), 10→0 is not an isolated prompt effect. **Devanagari drift:** under v1, 8/12 R explanations contained Devanagari despite the English instruction (0/12 for E and for C); under v2, 0/12. (Heuristic detector.)

Cross-model format compliance. Six more models ran on the same prompts under frozen v2 (Table 1). Phi-4-mini's inline FINAL lines (rejected by both endpoints) and R repetition to the token cap (6/12) make its C > E ordering a formatting artifact. Within Qwen3, instruction following broke before mathematics (36/36 invalid at 1.7B, 0/36 at 4B and 8B); without a compliance check, a multi-model comparison would mostly measure formatting.

Repeated *bhaag/bhinn* misreadings. Under v2, Qwen3-4B's only errors were two R items (AI-checked only): *bhaag dekar bhinn likho* ("divide, write as a fraction") was answered as a difference, and *kitna bhinn istemaal hua* ("what fraction was used") as the remaining count, as Llama-3.2-3B also did; *bhinn* also means "different". With $n = 2$, this is *consistent with* operation-word misreading, not evidence for it, and motivated the lexicon and keep/switch design.

6 Pilot Results

Each cohort was analysed under a pre-registration fixed before any of its outputs; Pilot-B is primary. The original cohort's plan was written with Pilot-B results known, so it is a descriptive replication. Each backbone ran once per cohort and completed every rendering (210/210; 72/72). The cohorts differ in items, review and C definition and are never pooled.

Backbone	E−R	E−C	R−C
Pilot-B (primary), 70 families			
Qwen3-4B	+8.6 [−1.4, +18.6] (9/3)	0.0 [−7.1, +7.1] (3/3)	−8.6 [−15.7, −1.4] (1/7)
Qwen3-8B	+7.1 [0.0, +14.3] (6/1)	0.0 [−4.3, +4.3] (1/1)	−7.1 [−14.3, 0.0] (1/6)
Llama-3.2-3B	+37.1 [+24.3, +50.0] (28/2)	+24.3 [+11.4, +37.1] (21/4)	−12.9 [−27.1, +1.4] (10/19)
Original cohort, 24 families			
Qwen3-4B	+8.3 [0.0, +20.8] (2/0)	0.0 [0.0, 0.0] (0/0)	−8.3 [−20.8, 0.0] (0/2)
Qwen3-8B	+4.2 [0.0, +12.5] (1/0)	0.0 [0.0, 0.0] (0/0)	−4.2 [−12.5, 0.0] (0/1)
Llama-3.2-3B	+25.0 [+4.2, +45.8] (7/1)	−4.2 [−16.7, +8.3] (1/2)	−29.2 [−45.8, −12.5] (0/7)

Table 3: Paired family differences (percentage points) with 95% intervals from 10,000 family-bootstrap resamples (seed 101), computed within each cohort. In parentheses: discordant families correct only in the first / only in the second form. Zero-width intervals at ceiling are not evidence of equivalence.

Romanized Hindi is the hard case. R is the lowest-scoring form for every backbone in both cohorts (Tables 2 and 3). In Pilot-B the deficit is modest for Qwen (8.6 and 7.1 points; intervals touch zero) and large for Llama (37.1 points, [+24.3, +50.0]); Qwen3-8B improves all forms but keeps the R deficit. The direction replicates for every model in the original cohort, where Qwen is near ceiling (E = C = 100%) and Llama's E−R deficit is 25.0 points [+4.2, +45.8]. Zero-width intervals at ceiling do not show equivalence.

Hinglish and English: equal aggregate accuracy for Qwen. Both Qwen models have equal observed aggregate accuracy on E and C in both cohorts (E−C = 0.0), but neither itemwise identity (Pilot-B discordances 3/3, 1/1) nor equivalence. C exceeds R; for Qwen3-4B in Pilot-B the R−C interval excludes zero. For Llama, C lies well below E in Pilot-B (+24.3 points) but not in the original cohort, so C's position is model- and cohort-dependent.

The operation-word hypothesis is not supported. In a pre-registered descriptive Pilot-B split by whether C keeps or switches R's Hindi operation word (or R has none), R accuracy is similar across strata for every model (keep/switch/none): Qwen3-4B 89/82/86%, Qwen3-8B 94/94/89%, Llama 67/59/51%. Many Llama R errors are arithmetic slips on simple items. In the original cohort, one Qwen3-4B R error (*Riya se 5 kam* "5 fewer than Riya" read as addition) is an isolated operation-word misreading.

Residual format effects. The numeric endpoint agrees with strict scoring on every Llama Pilot-B rendering and on all original-cohort renderings, which does not rule out failures both reject (Llama: 11 strict-invalid Pilot-B outputs, 7 truncated). For Qwen, the pre-registered secondary grammar rejects a *trailing* rupee symbol (`FINAL: 1575 ₹`) that strict scoring accepts, explaining every strict-correct/numeric-fail case (30 for Qwen3-4B, 36 for 8B); a post-hoc allowance restores agreement. Three of Qwen3-4B's four Pilot-B E failures are correct values with unlisted unit phrases (`11 years old`), and one C failure an unlisted spelling (`300 rupye`).

Audit and instruction following. Two AI models pre-tagged Qwen3-4B's Pilot-B outputs blind to each other ($\kappa = 0.94, 0.96$); a human checked all disagreements and 20 agreed rows (App. B). Of 18 strict failures, 6 are format-only; substantive errors concentrate in R (8, vs. 1 in E and 3 in C). English explanations fell from 70/70 (E) to 53/70 (R) and 59/70 (C).

7 Fine-tuning Follow-up

A separate pre-registration asked whether adding Hinglish to English training helps Hinglish more than adding Romanized Hindi. Three QLoRA arms of Qwen3-4B (E+E, E+R, E+C; identical English targets) were trained with three paired seeds, frozen by hash and evaluated once on 100 held-out families (App. C). The confirmatory E+C−E+R contrast on C was −4.3 points [−8.0, −0.7], negative for every seed: label *harm*. All arms lost 24–26 points on E against the unchanged model, with answers a third as long, so this tests a training recipe, not code-switching as such.

8 Conclusion

With the mathematics fixed, small open models were least accurate on Romanized Hindi, by a model-specific amount; formats need checking on excluded data, and naive Hinglish fine-tuning did not help.

Limitations

- **Diagnostic scale.** With 70 and 24 families per cohort, intervals are wide and the vocabulary strata are sparse and descriptive. The original cohort is near ceiling for Qwen, and its analysis plan was written after Pilot-B results were known.
- **Thin development evidence.** §5 rests on 12 unreviewed fixture families, and the prompt was revised after seeing their outputs; no pilot item was used to design the prompt.
- **Limited language review.** Pilot-B, the test set and the lexicon each had one bilingual reviewer, who is a team member; the original cohort had one human language reviewer plus an adjudicator. Part of the lexicon was AI-assisted, and *anupat* is unconfirmed. Naturalness judgments reflect few speakers' register.
- **AI-authored data.** Items share LLM drafting styles. Claude both drafted and solved Pilot-B. The GPT/Claude repair contrast is confounded with split and prompt.
- **Different C definitions.** The original cohort and Pilot-B define C differently, which prevents a simple pooled C estimate.
- **Models.** All backbones are small, quantized, greedy and non-thinking. Admitting models by format compliance excludes families whose formatting the scorer does not accept.
- **No script control.** There is no Devanagari arm, so an E–R gap mixes script, language, register and spelling. C $\approx$ E with R $<$ E would suggest vocabulary but would not identify it. Each R rendering is one of many possible spellings.
- **Fine-tuning recipe.** The follow-up trains on short reference solutions, which degraded every arm; it cannot separate a code-switching effect from this recipe, used one 4B backbone, and its interval is conditional on the trained checkpoints, ignoring seed variance. The test set's R and C were GPT-drafted and repaired by one reviewer.
- **Contamination.** The items are new, but the models' pretraining data cannot be audited.

Ethics Statement

Consent and credit. Reviewers received a consent note covering the task, data use and release; all reviewers consented to release under CC BY 4.0. Reviewers will be credited in the camera-ready acknowledgments or as authors, as appropriate and according to their stated preference (the adjudicator declined credit). Real identities stay out of shared ledgers.

AI-assistance disclosure. (1) *AI-drafted items:* renderings were drafted by an LLM (Claude for Pilot-B and the training split, GPT for test and development); for the original pilot, the English statements are author-written and R and C were LLM-translated. The families were then checked against code-computed gold and reviewed by humans; Pilot-B was also verified blind by two LLM solvers. (2) *AI-assisted human review:* humans made every acceptance decision, but some lexicon entries and some text edits were drafted by AI before reviewers confirmed them (§3). In the original cohort, one reviewer's AI-prefilled, unverified stage-2 judgments were excluded. (3) *Writing and coding:* AI assistants helped write the evaluation and analysis code, the research notes and this manuscript, and recorded protocol decisions under the authors' delegation. The authors reviewed all of this and take responsibility for it Systems used: Claude Opus 5.5 (Anthropic) and GPT-6 Astra and GPT-6.1 Sol (OpenAI, via the Codex CLI).

Data. The items are invented everyday word problems and are not expected to contain personal information (they use invented first names; an automated scan found no contact details). Romanized Hindi varies by region and register, and our renderings reflect few speakers. Data are released under CC BY 4.0 and code under MIT. The test set was used once, for the confirmatory fine-tuning evaluation, and will be released with the camera-ready version.

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A Frozen prompt

System message used for every form and model (`test1-english-final-v2`):

> `Solve the problem. Give a short explanation in English, even if the question is not in English. End with exactly one line: FINAL: <number or fraction>, followed by its unit in English only if the answer has a unit; write nothing after a pure number or fraction.`

The user message is the question text alone. Decoding is greedy and non-thinking, with a 1,024-token output cap.

B Audit of Qwen3-4B Pilot-B outputs

Primary category per output (first applicable in the pre-registered order; 70 per form). Labels are the agreed AI pre-tag (186 outputs) or the human label (24 outputs: the 4 tagger disagreements and 20 random agreed rows). One human label, "ambiguous" (PB033-R, flagged as an ambiguous question), is mapped to *unresolved*.

Category	E	R	C
no_error	66	47	56
format_only	3	2	1
language_drift	0	13	10
wrong_relation_operation	1	4	2
misread_quantity	0	2	0
arithmetic_slip	0	1	1
unresolved	0	1	0
English-compliant explanation	70	53	59

Exploratory (not pre-registered): by a heuristic count, Llama explained 60/70 Pilot-B R outputs in Romanized-Hindi prose. On Llama's Pilot-B C outputs, Romanized-Hindi-prose explanations were correct 13/27 against 36/43 for English explanations, an association confounded by item difficulty.

C Fine-tuning follow-up: design and results

Design. Qwen3-4B (same revision), NF4 QLoRA, rank 16, $\alpha = 32$, 60 updates of 16 examples (three passes over 160 training families), learning rate 10^{-4} with six warm-up updates and cosine decay to 10^{-5}. Each arm pairs every English item with its rendering in the named form; all arms supervise the same English solution. Seeds 11/22/33 fix initialization and data order, shared across arms. The 40-family DEV split was used only for sanity checks, never for selection. Infrastructure failures allowed one identical restart; none was needed (9/9 runs valid at the first attempt). The adapters, code and baseline were frozen by SHA-256 before test access. The test set has 100 families (300 renderings per checkpoint), scored with the frozen strict scorer and unchanged prompt. **Decision rule** on the 95% paired family bootstrap of E+C−E+R on C (seed-averaged; 10,000 resamples): harm if $U < 0$; informative null if $L \le 0 \le U < 5$; inconclusive if $L \le 0$ and $U \ge 5$; support if $L > 0$ (useful support if $L > 5$); a degenerate interval overrides all labels. Two evaluation interruptions were resumed from stored responses (one generation each, none regenerated); an independent reimplementation reproduced every number exactly.

Strict accuracy (%)	E	R	C
Unchanged Qwen3-4B	85.0	63.0	66.0
E+E (mean of 3 seeds)	61.3	40.3	43.0
E+R (mean of 3 seeds)	59.3	38.0	47.0
E+C (mean of 3 seeds)	58.7	41.3	42.7

Contrast (points)	Est. [95% bootstrap]
E+C−E+R on C (primary)	−4.3 [−8.0, −0.7]
E+C−E+E on E	−2.7 [−7.0, +1.3]
E+C−unchanged on E	−26.3 [−34.3, −19.0]
E+R−E+E on R	−2.3 [−9.3, +4.3]
E+C−E+R on R	+3.3 [−2.0, +8.7]

Per-seed E+C−E+R on C: −3, −1, −9 points. The secondary numeric endpoint gives −5.7 [−11.0, −0.7]. Rows other than the first are descriptive. Median output length is about 42 tokens for trained arms against 111 for the unchanged model; in an exploratory decomposition, numerically wrong answers account for most of the trained arms' deficit against the unchanged model (13–22 points per form), and rejected units for the rest (3–11 points). On these test items the unchanged model again scores lowest on R (63%), but C (66%) lies far below E (85%), unlike Pilot-B; this set was GPT-drafted and its R needed heavy repair (§3).